Darke Theory: A Pre-Geometric Bridge Between Quantum Gravity and General Relativity

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Abstract

This paper presents Darke Theory as a unifying framework for quantum gravity and general relativity. Its central claim is that spacetime, geometry, and physical law are not primitive, but emerge only once coherence exceeds a critical threshold sufficient for stable closure. On this view, quantum mechanics and general relativity are not competing foundations, but threshold-separated regimes of one deeper pre-geometric structure. Below the coherence threshold, geometric notions such as distance, duration, locality, and curvature are undefined, and the resulting regime is the one ordinarily described quantum mechanically. Above that threshold, geometry becomes admissible and an effective relativistic curvature regime appears.

Within this structure, the relativistic closure–curvature bridge provides an explicit post-threshold field system linking metric, coherence, and matter, together with a theorem chain establishing Lorentzian regime emergence, causal hyperbolicity, matter as persistent topological coherence, gravitational consistency, weak-field reduction, and vacuum stability. The DTFE sector then supplies the first concrete matter-sector candidate for that bridge: a frozen nonlinear parent action with a live topological branch and strong classical validation.

The present paper sharpens and partially closes the unification claim by deriving the matter–geometry seam rather than merely naming it. It identifies the canonical induced post-threshold matter field obtained from a nontrivial DTFE branch, proves exact transfer of topological nontriviality into the bridge sector, proves finite bridge energy under standard post-threshold regularity hypotheses, derives the corresponding induced bridge stress tensor, and shows that the induced sector enters the post-threshold bridge equation as a finite persistent curvature source. The unification proposed here is therefore not a later reconciliation of two independent theories, but the derivation of both regimes from one underlying coherence structure, with DTFE furnishing the first concrete executable route by which the bridge may be materially realised in derived form inside the post-threshold regime.

1 Introduction

The question of quantum gravity is usually framed as a conflict between two successful but structurally incompatible theories. General relativity presupposes a geometric spacetime and describes gravity as curvature of that spacetime. Quantum mechanics presupposes a formal state space with unitary evolution and exhibits nonlocal correlation structures that do not sit naturally inside a primitive geometric ontology. Standard approaches attempt to quantise gravity, geometrise quantum structure, or construct a higher formalism in which both appear as limiting cases.

Darke Theory approaches the problem differently. It does not begin with spacetime, metric distance, temporal succession, or already-furnished law. Instead it begins with a pre-geometric regime of relational coherence and asks under what conditions stable geometric reference becomes possible at all. In that framework, geometry is not a primitive stage on which physics unfolds. Geometry is activated only after coherence exceeds a threshold sufficient for stable closure. Matter is not primitive substance but stable topological coherence. Gravity is not fundamental interaction in the usual sense, but the geometric projection of coherence gradients once geometry has become admissible.

This changes the unification problem fundamentally. If spacetime is emergent, then the problem

is not to force quantum states into a pre-existing geometry. It is to show how the same underlying structure gives rise to both the near-threshold regime ordinarily described by quantum mechanics and the post-threshold regime ordinarily described by general relativity. That is the burden of the present paper.

The paper does not claim that every physical sector is now derived in exhaustive final form. It does not claim that all particle sectors are already identified, nor that every quantitative phenomenological consequence has already been extracted. Its claim is narrower and more exact. The architecture required for unification is now explicit at four levels:

(i) a pre-geometric ontological foundation in which quantum and relativistic regimes are downstream of coherence closure;

(ii) a post-threshold relativistic bridge with explicit metric, coherence, and matter sectors;

(iii) a first executable field-theoretic matter sector, DTFE, that may furnish the first concrete realisation of post-threshold matter within that bridge;

(iv) a derived matter-sector transfer showing how a nontrivial DTFE branch induces a finite-energy nontrivial bridge matter sector and enters the post-threshold curvature source under explicit regularity conditions.

The aim of the paper is therefore to state the unification problem in its now-correct and partially closed form. The central issue is no longer whether one can imagine a conceptual bridge between quantum mechanics and general relativity. The central issue is whether the existing bridge can be materially realised by a concrete topological matter sector in the post-threshold regime while remaining faithful to the pre-geometric origin of the theory. The present paper shows that this can be done in derived form for the DTFE matter candidate under explicit post-threshold regularity hypotheses.

2 Exact claim boundary

The paper depends on a strict claim boundary. It is therefore necessary to say clearly what is and is not being asserted.

2.1 What is claimed

Proposition 2.1 (Current claim boundary)**.** *Darke Theory now contains, in explicit form:*

(a) a pre-geometric ontological foundation in which spacetime and geometry arise only after closure threshold crossing;

(b) an interpretation of quantum mechanics as a near-threshold pre-geometric regime of unresolved coherence;

(c) an explicit post-threshold relativistic bridge with primitive fields $(g_{\mu\nu}, \Phi, \psi)$ *and a theorem chain establishing Lorentzian regime emergence, causal hyperbolicity, matter as persistent topological coherence, gravitational consistency, weak-field reduction, and vacuum stability;*

(d) a concrete executable field-theoretic sector, DTFE, with a live topological branch and strong classical validation;

(e) a canonical induced bridge field obtained directly from a nontrivial DTFE branch;

(f) an exact topological transfer statement identifying the induced bridge topological charge with the DTFE degree;

(g) a finite-energy transfer theorem under explicit post-threshold regularity assumptions;

(h) a derived curvature-sourcing statement showing that the induced DTFE sector enters the bridge equation as a finite persistent matter source under those same assumptions.

2.2 What is not claimed

The paper does not claim:

(a) that every sector of the full theory has already been derived in final unconditional form without remaining regularity assumptions;

(b) that the complete particle catalogue has already been derived;

(c) that every phenomenological consequence has already been worked out;

(d) that the full quantum completion of DTFE, or the complete nonradial closure of every DTFE sector, is already required or already completed for the present unification claim.

Remark 2.2. *The paper is therefore neither a speculative manifesto nor a claim of total closure across every possible sector. It is a theory paper whose purpose is to identify the exact point at which unification becomes materially derivable, to close the matter–geometry seam in the strongest honest form presently supported, and to state the remaining burden precisely where it still exists.*

3 Pre-geometric foundation

Darke Theory begins from a strict ontological boundary: spacetime, geometry, and physical law are not primitive. They are emergent features of a deeper pre-geometric regime of relational coherence. A dimensionless coherence functional $\Phi[C]$ is defined over relational configuration space C, and geometry is said to exist if and only if coherence exceeds a critical threshold:

$$\exists G \iff \Phi[C] > \Phi_{\text{crit}}. \tag{1}$$

In the closed regime, coherence is locally stationary,

$$\delta\Phi = 0, \tag{2}$$

allowing persistent reference, structural memory, and the emergence of classical spacetime dynamics.

This is the foundational move on which the whole unification project depends. The Planck scale is reinterpreted not as a smallest unit of spacetime, but as the regime boundary at which geometry first becomes well-defined. Below that boundary, length, duration, locality, and curvature are not merely small or noisy; they are undefined and inapplicable.

Remark 3.1. *This is what allows Darke Theory to treat quantum gravity and general relativity as two emergent regimes of one deeper structure. If geometry is already presupposed at the base, the theory has lost the unification game before it begins.*

4 Quantum regime below closure threshold

Within this framework, Darke Theory does not alter the formalism of quantum mechanics. It reinterprets its ontology. Quantum states are understood as pre-geometric coherence structures that have not yet achieved closure sufficient to define spacetime relations. Superposition is therefore treated as unresolved relational possibility rather than as contradictory physical actuality.

Measurement is interpreted as structural resolution. When coherence commits to a closed configuration, geometric reference becomes possible. Because closure has not yet occurred prior to measurement, unitary evolution and reversibility are preserved at the quantum level.

This interpretation also gives a direct account of Bell nonlocality. Quantum correlations are established prior to the emergence of spacetime locality and are therefore not constrained by geometric separation. Once closure occurs, those correlations are inherited by geometric degrees of freedom without requiring superluminal signalling or preferred frames.

Thus the quantum regime is not fundamental in the ontological sense. It is the near-threshold regime in which coherence is partially unconstrained and geometric description has not yet become admissible.

5 Post-threshold relativistic bridge

The relativistic closure–curvature bridge provides the post-threshold field system in which effective geometry, matter, and curvature become explicit. The primitive post-threshold fields are

$$(g_{\mu\nu}, \Phi, \psi), \tag{3}$$

where $g_{\mu\nu}$ is a smooth symmetric nondegenerate tensor, Φ is a coherence-density field, and ψ is a concrete closure-matter field.

The bridge concerns only the post-threshold regime

$$\Phi(x) > \Phi_{\text{crit}} \qquad \forall x \in M. \tag{4}$$

Geometry is activated only there. A geometry-activation coupling

$$F(\Phi) > 0 \tag{5}$$

enters the metric sector, and the bridge is built under explicit positivity, hyperbolicity, and admissibility conditions.

Its purpose is exact: to build a relativistic bridge from post-threshold closure to an effective curvature and matter regime. In full form, the bridge is designed to establish:

1. geometry activates only above a closure threshold;
2. admissibility is represented by a real functional on the field space;
3. curvature is sourced by coherence gradients, matter sectors, and admissibility stress;
4. causal structure arises from hyperbolicity and is physically filtered by closure balance;
5. stable topological coherence sectors act as matter-energy concentrations;
6. the coupled metric–coherence–matter system is gravitationally consistent on shell;

7. the post-threshold vacuum is linearly stable under explicit positivity conditions;
8. a homogeneous cosmological sector exists without taking the standard Friedmann equation as primitive.

The final bridge theorem states that if $(g_{\mu\nu}, \Phi, \psi)$ satisfy post-threshold closure, geometric activation, positive kinetic structure, stationarity of the full bridge action, hyperbolicity, Lorentzian signature, and finite admissibility, then:

(a) the post-threshold regime admits an effective Lorentzian geometry;

(b) curvature is sourced by coherence gradients, matter sectors, and admissibility stress through

$$F(\Phi)G_{\mu\nu} = \kappa T^{\text{eff}}_{\mu\nu}; \tag{6}$$

(c) finite-energy nontrivial topological sectors of ψ act as matter-energy concentrations;

(d) physical propagation lies in the hyperbolic cone filtered by closure-balance admissibility;

(e) the coupled system is covariantly consistent on shell;

(f) the post-threshold vacuum is linearly stable under the stated positivity conditions.

This theorem chain is the bridge itself.

6 Matter as persistent closed coherence

The bridge does not treat matter as primitive inserted substance. It defines a concrete closure-matter sector by the Lagrangian

$$L_m = -\frac{1}{2}K(\Phi)\, g^{\mu\nu}\nabla_\mu\bar{\psi}\,\nabla_\nu\psi - U(\psi, \Phi), \tag{7}$$

with

$$K(\Phi) > 0, \qquad U(\psi, \Phi) = \lambda_\psi\big(|\psi|^2 - v(\Phi)^2\big)^2. \tag{8}$$

A finite-energy topological sector is defined by

$$E[\psi] < \infty, \qquad Q[\psi] \neq 0, \tag{9}$$

where $Q[\psi]$ is the topological charge.

The matter theorem of the bridge then states that if

$$K(\Phi) > 0, \qquad U(\psi, \Phi) \geq 0, \qquad E[\psi] < \infty, \qquad Q[\psi] \neq 0, \tag{10}$$

then ψ defines a nontrivial persistent topological coherence sector whose stress-energy acts as a persistent curvature source. Such sectors are macroscopically interpretable as matter-energy concentrations.

This is decisive. It means the bridge is not merely saying that matter might someday appear. It already contains an explicit theorem-level criterion for what matter is in the post-threshold regime.

7 DTFE as the first concrete matter sector

This is the point at which DTFE enters.

DTFE is not the whole of Darke Theory. It is a frozen nonlinear parent-action theory that may provide the first serious executable field-theoretic route by which post-threshold closure structure is realised explicitly rather than merely identified conceptually. Its frozen parent field is an $SU(2)$ field

$$U(x) \in SU(2), \qquad L_\mu := U^\dagger \partial_\mu U, \tag{11}$$

with parent action

$$S_{\mathrm{DTFE}}[U] = \int d^4x \left[-\frac{1}{2}\mathrm{Tr}(L_\mu L^\mu) + \frac{1}{16}\mathrm{Tr}([L_\mu, L_\nu][L^\mu, L^\nu]) \right]. \tag{12}$$

Under the hedgehog ansatz

$$U(x) = \cos F(R)\, \mathbf{1} + i\, \hat{x} \cdot \tau \, \sin F(R), \qquad R = |x|, \tag{13}$$

the system reduces to a live radial branch with

$$F(0) = \pi, \qquad F(\infty) = 0. \tag{14}$$

The field-equation paper is exact about the current status. It does not claim that the final particle-identification step has already been closed. It does claim that DTFE is now frozen, executable, and strongly validated at the classical parent-action level, and that it may be the first frozen field-theoretic route by which post-threshold closure structure begins to be realised explicitly.

This is what makes DTFE central to the unification programme. The bridge already has a matter theorem in terms of finite-energy topological coherence. DTFE already has a live topological branch and strong classical validation. The remaining problem is therefore no longer whether Darke Theory can talk about matter in principle. The problem is whether the DTFE topological sector can be identified as the first concrete realisation of the bridge matter sector. The present paper shows that this can be done in derived form under explicit post-threshold regularity assumptions.

7.1 Canonical DTFE-to-bridge induced field

The first task is to construct the induced bridge matter field from a finite-energy nontrivial DTFE branch in a form that is canonical up to the expected internal redundancy.

Setting

Let

$$M_{\mathrm{post}} := \{x \in M : \Phi(x) > \Phi_{\mathrm{crit}}\} \tag{15}$$

denote the post-threshold region, where geometric description is admissible. In Darke Theory, this is the only regime in which a geometric matter field can be meaningfully defined. Below threshold, geometry is not yet admissible, and geometric matter variables are therefore not primitive objects of the theory. Post-threshold matter must accordingly be induced from closure-bearing structure rather than assumed independently.

Let

$$U_B : M_{\mathrm{post}} \to SU(2) \tag{16}$$

be a finite-energy nontrivial DTFE branch. In the frozen DTFE parent-action sector, such a branch is the concrete field-theoretic realisation of stable topological closure presently available. The task is to construct from U_B a post-threshold bridge field ψ_{U_B} that lies in the bridge matter class and is not chosen ad hoc.

Reference doublet construction

Fix a unit reference doublet

$$\xi_0 \in \mathbb{C}^2, \qquad \|\xi_0\| = 1. \tag{17}$$

Define the associated complex doublet field

$$z_B(x) := U_B(x)\,\xi_0. \tag{18}$$

Since $U_B(x) \in SU(2)$ acts unitarily on $\mathbb{C}^2$, one has

$$\|z_B(x)\| = \|\xi_0\| = 1 \tag{19}$$

for all $x \in M_{\text{post}}$. Thus z_B is a unit-norm complex doublet representative canonically induced by the DTFE branch once a reference internal direction is fixed.

The bridge matter field is then defined by

$$\psi_{U_B}(x) := v(\Phi(x))\,z_B(x) = v(\Phi(x))\,U_B(x)\xi_0. \tag{20}$$

Why this is the correct induced field

This construction is not introduced as an arbitrary convenience. It is forced by the structural roles already fixed in the theory.

First, the induced object must live in the post-threshold regime only. The factor $v(\Phi)$ ensures that the induced field is tied directly to the same coherence-controlled vacuum structure that appears in the bridge matter sector. It therefore vanishes as an autonomous geometric matter field when the threshold condition fails.

Secondly, the induced object must carry the nontrivial topological content of the DTFE branch into the bridge matter functional with the least possible additional structure. The map

$$U_B \mapsto U_B\xi_0 \tag{21}$$

is exactly the minimal fundamental-complex representation obtained from the $SU(2)$ branch itself. It does not introduce an independent matter field by hand. It reads one directly off the DTFE configuration.

Thirdly, the induced object must be compatible with the bridge matter Lagrangian. A complex doublet field with coherence-dependent vacuum scale is precisely the form required for the post-threshold matter sector already present in the relativistic closure–curvature bridge.

For these reasons, ψ_{U_B} is the minimal post-threshold carrier of DTFE topological content into the bridge matter class.

Definition 7.1 (Canonical induced bridge field)**.** *Let $U_B : M_{\text{post}} \to SU(2)$ be a finite-energy nontrivial DTFE branch, and let $\xi_0 \in \mathbb{C}^2$ satisfy $\|\xi_0\| = 1$. The canonical induced bridge field associated to U_B is*

$$\psi_{U_B}(x) := v(\Phi(x))\,U_B(x)\xi_0. \tag{22}$$

Proposition 7.2 (Well-definedness of the induced field)**.** *The field ψ_{U_B} is well-defined on M_{post}, smooth whenever U_B and Φ are smooth, and transforms only by the expected internal redundancy under change of unit reference doublet.*

Proof. Since U_B is $SU(2)$-valued and $\xi_0 \in \mathbb{C}^2$, the product $U_B\xi_0$ is a globally defined complex doublet on the post-threshold region. If U_B and Φ are smooth, then so is $\psi_{U_B} = v(\Phi)U_B\xi_0$.

Now let ξ_1 be another unit reference doublet. Since $SU(2)$ acts transitively on the unit sphere in $\mathbb{C}^2$, there exists $S \in SU(2)$ such that

$$\xi_1 = S\xi_0. \tag{23}$$

Then

$$U_B\xi_1 = U_BS\xi_0. \tag{24}$$

This changes the representative of the induced doublet but not the existence of the induced bridge sector itself. The resulting ambiguity is exactly the expected internal redundancy of the doublet description, not an ambiguity in whether the transfer exists. □

Theorem 7.3 (Canonical induced-field theorem)**.** *Let $U_B : M_{\text{post}} \to SU(2)$ be a finite-energy nontrivial DTFE branch. Then the assignment*

$$U_B \longmapsto \psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{25}$$

defines a canonical post-threshold bridge matter field up to the natural internal redundancy determined by the choice of unit reference doublet ξ_0.

Proof. The branch U_B is the given post-threshold topological closure object. Any bridge matter field induced from it must satisfy three conditions:

1. it must be defined only where geometry is admissible;
2. it must be constructed directly from the DTFE branch rather than added independently;
3. it must belong to the field class accepted by the bridge matter functional.

The field

$$\psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{26}$$

satisfies (i) because $v(\Phi)$ is a post-threshold coherence-controlled scale; it satisfies (ii) because the only nontrivial field content is inherited directly from U_B; and it satisfies (iii) because it is a coherence-scaled complex doublet of the exact kind admitted by the bridge matter sector. Any alternative induced field carrying the same content with less structure would have to omit either the post-threshold scaling factor or the direct extraction from the $SU(2)$ branch, and would therefore fail either geometric threshold fidelity or bridge compatibility.

Hence the induced field is canonical up to the expected internal redundancy associated with the choice of ξ_0. □

Remark 7.4. *The canonical induced-field theorem does not yet by itself show that the induced field belongs to the bridge matter class. That burden is met by the derived transfer theorems below.*

7.2 Derived DTFE-to-bridge transfer

The unification seam can now be sharpened further. The canonical induced bridge field

$$\psi_{U_B}(x) = v(\Phi(x))\, U_B(x)\xi_0, \qquad \|\xi_0\| = 1, \tag{27}$$

is no longer merely a formally compatible candidate. The present subsection records the exact derived content that follows from this construction together with the frozen DTFE branch and the post-threshold bridge structure.

Normalised induced field and bridge charge

Because $U_B(x) \in SU(2)$ acts unitarily on $\mathbb{C}^2$, one has

$$|U_B(x)\xi_0| = 1 \tag{28}$$

for all $x \in M_{\text{post}}$. Since the bridge vacuum scale $v(\Phi)$ is positive in the post-threshold regime, the induced field has nonvanishing norm there and therefore admits a canonical normalisation

$$\widehat{\psi}_{U_B}(x) := \frac{\psi_{U_B}(x)}{|\psi_{U_B}(x)|} = U_B(x)\xi_0. \tag{29}$$

This is the decisive simplification. The topological content of the induced bridge field is therefore entirely carried by the S^3-valued field $U_B\xi_0$.

Definition 7.5 (Bridge topological charge induced from the normalised field)**.** *Let Σ be a spatial post-threshold slice on which the finite-energy DTFE branch approaches a constant vacuum value at spatial infinity. After one-point compactification, $\Sigma^+ \cong S^3$. Define the induced bridge topological charge of ψ_{U_B} by*

$$Q_{\text{br}}[\psi_{U_B}] := \deg(\widehat{\psi}_{U_B}) = \deg(U_B\xi_0). \tag{30}$$

Remark 7.6. *This definition is not imposed arbitrarily. The bridge matter theorem requires a finite-energy nontrivial topological sector of ψ, while the canonical induced field already identifies a normalised S^3-valued representative. The degree of that representative is therefore the minimal bridge charge compatible with the existing bridge matter theorem and the DTFE construction.*

Proposition 7.7 (Topological transfer lemma)**.** *Let $U_B : \Sigma^+ \to SU(2)$ be a finite-energy nontrivial DTFE branch with vacuum limit at spatial infinity. Fix a unit reference doublet $\xi_0 \in \mathbb{C}^2$. Then the map*

$$\Theta_{\xi_0} : SU(2) \to S^3, \qquad \Theta_{\xi_0}(U) := U\xi_0, \tag{31}$$

is a diffeomorphism. Consequently,

$$\deg(U_B\xi_0) = \deg(U_B). \tag{32}$$

Proof. Choose coordinates so that $\xi_0 = e_1 = (1,0)^T$. Every element of $SU(2)$ has the form

$$U = \begin{pmatrix} a & b \\ -\bar{b} & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1. \tag{33}$$

Then

$$Ue_1 = \begin{pmatrix} a \\ -\bar{b} \end{pmatrix}, \tag{34}$$

which ranges over the full unit sphere in $\mathbb{C}^2 \cong \mathbb{R}^4$, that is, over S^3. Moreover, the first column determines the second uniquely by the $SU(2)$ condition, so the map is bijective and smooth with smooth inverse. Therefore Θ_{ξ_0} is a diffeomorphism. Degree is invariant under postcomposition by diffeomorphism, hence

$$\deg(U_B\xi_0) = \deg(U_B).$$

□

Theorem 7.8 (Exact topological transfer theorem)**.** *Let U_B be a finite-energy nontrivial DTFE branch on a post-threshold spatial slice with vacuum limit at infinity, and let*

$$\psi_{U_B} = v(\Phi)\, U_B\xi_0 \tag{35}$$

be the canonical induced bridge field. Then

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(U_B). \tag{36}$$

In particular, if the DTFE branch is topologically nontrivial, then

$$Q_{\mathrm{br}}[\psi_{U_B}] \neq 0. \tag{37}$$

Proof. By definition of the normalised induced field,

$$\widehat{\psi}_{U_B} = U_B\xi_0.$$

Hence

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(\widehat{\psi}_{U_B}) = \deg(U_B\xi_0).$$

By the previous proposition, $\deg(U_B\xi_0) = \deg(U_B)$. Therefore

$$Q_{\mathrm{br}}[\psi_{U_B}] = \deg(U_B).$$

If U_B is topologically nontrivial, the degree is nonzero, so the induced bridge charge is nonzero as well. □

Finite-energy transfer

The bridge matter functional is

$$L_m = -\frac{1}{2}K(\Phi)\, g^{\mu\nu}\nabla_\mu\bar{\psi}\,\nabla_\nu\psi - U(\psi,\Phi), \qquad U(\psi,\Phi) = \lambda_\psi\big(|\psi|^2 - v(\Phi)^2\big)^2. \tag{38}$$

For the canonical induced field,

$$|\psi_{U_B}|^2 = |v(\Phi)|^2\,|U_B\xi_0|^2 = v(\Phi)^2, \tag{39}$$

so the potential term vanishes identically:

$$U(\psi_{U_B},\Phi) = 0. \tag{40}$$

Differentiating,

$$\nabla_\mu\psi_{U_B} = (\partial_\mu v(\Phi))\, U_B\xi_0 + v(\Phi)\,(\nabla_\mu U_B)\xi_0. \tag{41}$$

Using unitarity and $\|\xi_0\| = 1$,

$$|\nabla\psi_{U_B}|^2 \leq 2|\nabla v(\Phi)|^2 + 2|v(\Phi)|^2\,\|\nabla U_B\|^2. \tag{42}$$

Theorem 7.9 (Finite-energy transfer theorem)**.** *Assume, in addition to the hypotheses above, that on the post-threshold region:*

1. *$K(\Phi)$ is bounded above and below by positive constants;*
2. *$v(\Phi)$ is bounded;*
3. *$\nabla v(\Phi) \in L^2$ on the post-threshold spatial slice;*
4. *the frozen DTFE branch has finite gradient content on that slice, equivalently*

$$\int \|\nabla U_B\|^2 \, d\mu_g < \infty. \tag{43}$$

Then the induced bridge field has finite bridge energy:

$$E[\psi_{U_B}] < \infty. \tag{44}$$

Proof. The potential term vanishes identically. For the kinetic term, the estimate above gives

$$K(\Phi)|\nabla \psi_{U_B}|^2 \leq 2K_+|\nabla v(\Phi)|^2 + 2K_+ v_+^2 \|\nabla U_B\|^2, \tag{45}$$

where K_+ and v_+ are upper bounds for $K(\Phi)$ and $|v(\Phi)|$. Both terms on the right are integrable by assumptions (3) and (4). Hence the bridge energy is finite. □

Remark 7.10. *The finite-gradient assumption on U_B is the bridge-side regularity translation of the finite gradient content suggested by the frozen DTFE parent-action energy. In the present paper it is stated explicitly as a post-threshold regularity hypothesis. Deriving it directly from the full parent-action energy structure is part of the remaining strengthening programme, not a defect in the derived bridge logic established here.*

Derived curvature sourcing

The bridge already proves that finite-energy nontrivial topological sectors of ψ act as matter-energy concentrations and source curvature through

$$F(\Phi)G_{\mu\nu} = \kappa T^{\text{eff}}_{\mu\nu}. \tag{46}$$

We may therefore identify the induced DTFE contribution directly from the bridge matter Lagrangian.

Proposition 7.11 (Induced bridge stress tensor)**.** *For the canonical induced field ψ_{U_B}, the bridge matter stress tensor is*

$$T^{(\psi_{U_B})}_{\mu\nu} = K(\Phi)\Big(\nabla_{(\mu}\bar{\psi}_{U_B}\nabla_{\nu)}\psi_{U_B} - \frac{1}{2}g_{\mu\nu}\, g^{\alpha\beta}\nabla_\alpha\bar{\psi}_{U_B}\nabla_\beta\psi_{U_B}\Big), \tag{47}$$

since the potential term vanishes identically on ψ_{U_B}.

Proof. This is the standard stress tensor obtained by variation of the bridge matter Lagrangian with respect to the metric. Because

$$U(\psi_{U_B}, \Phi) = 0,$$

no potential contribution remains. □

Theorem 7.12 (Derived curvature-sourcing theorem)**.** *Under the hypotheses of the finite-energy transfer theorem and the exact topological transfer theorem, the canonical induced field ψ_{U_B} defines a finite-energy nontrivial topological bridge matter sector. Consequently it contributes a finite persistent source term to the bridge equation, and the post-threshold field equations take the form*

$$F(\Phi)G_{\mu\nu} = \kappa\Big(T^{(\Phi)}_{\mu\nu} + T^{(\psi_{U_B})}_{\mu\nu} + T^{(\mathrm{adm})}_{\mu\nu}\Big). \tag{48}$$

Proof. By the finite-energy transfer theorem,

$$E[\psi_{U_B}] < \infty.$$

By the exact topological transfer theorem,

$$Q_{\mathrm{br}}[\psi_{U_B}] \neq 0$$

whenever the DTFE branch is topologically nontrivial. The bridge matter theorem then applies directly: finite-energy nontrivial topological sectors of ψ define persistent matter-energy concentrations and therefore contribute to the effective curvature source. Substituting the explicit induced matter stress tensor gives the stated sourced bridge equation. □

Corollary 7.13 (Closed matter–geometry seam under post-threshold regularity)**.** *Let U_B be a finite-energy nontrivial DTFE branch with vacuum limit at infinity, and let*

$$\psi_{U_B} = v(\Phi)\, U_B \xi_0 \tag{49}$$

be the canonical induced bridge field. Under the standard post-threshold regularity hypotheses used above, the DTFE branch determines:

1. *a canonically induced bridge matter field;*
2. *a nontrivial bridge topological charge equal to the DTFE degree;*
3. *finite bridge matter energy;*
4. *a finite persistent curvature source in the post-threshold bridge equations.*

Therefore the DTFE branch closes the matter–geometry seam in derived form within the post-threshold bridge regime under the stated regularity assumptions.

8 The unification problem in its now-correct form

At this stage the unification problem is sharply defined.

Proposition 8.1 (Current unification seam)**.** *The remaining central task in Darke Theory quantum gravity–general relativity unification is no longer to invent a bridge between DTFE and the post-threshold relativistic sector, but to extend, relax, or remove the explicit post-threshold regularity assumptions under which that bridge is now materially realised.*

This remaining burden can now be stated as a concrete strengthening programme rather than as a vague reconciliation slogan.

8.1 Strengthening of the finite-energy transfer

One may now seek to derive the finite gradient content of the induced DTFE sector directly from the full parent-action energy structure rather than assuming it in bridge form. This would further compress the remaining regularity burden.

8.2 Extension beyond the present post-threshold regularity class

One may seek to weaken the boundedness and integrability hypotheses on $K(\Phi)$, $v(\Phi)$, and $\nabla v(\Phi)$ while preserving the same derived transfer result. This is now a technical strengthening question, not the original vague unification seam.

8.3 Expanded weak-field and phenomenological extraction

Since the induced matter sector now enters the post-threshold curvature source in derived form, the next natural task is to extract explicit weak-field corrections and first phenomenological consequences from the sourced bridge equations.

8.4 Beyond the present matter candidate

The present paper identifies DTFE as the first concrete executable matter-sector route into the bridge. It does not assert that no other matter-sector routes exist. Future work may enlarge the matter catalogue without weakening the derived unification result already established for the DTFE branch.

9 Why the bridge is attainable

The unification is attainable because the bridge is no longer hypothetical. The required levels are already present.

First, the pre-geometric foundation already exists. Darke Theory states that quantum mechanics and general relativity are emergent regimes of one deeper structure, that matter is stable topological coherence, that gravity is the projection of coherence gradients into geometry, and that the Planck scale is a threshold of geometric admissibility rather than a fundamental geometric unit.

Secondly, the relativistic bridge already exists. The bridge theorem explicitly provides post-threshold metric, coherence, and matter fields together with Lorentzian regime emergence, persistent topological matter sectors, gravitational consistency, weak-field reduction, and vacuum stability.

Thirdly, the first executable matter candidate already exists. DTFE is a frozen nonlinear parent-action theory with a live topological branch and strong classical validation.

Fourthly, the exact transfer seam is now explicit in derived form. The paper states the canonical induced bridge field, proves exact topological transfer, proves finite bridge energy under explicit regularity conditions, and derives the corresponding induced curvature source.

So the bridge is attainable not because it can be imagined, but because its architecture is already present and its central matter–geometry seam is now materially realised inside the post-threshold regime.

Remark 9.1. *This is the correct sense in which Darke Theory is now near quantum gravity–general relativity unification in the strong technical sense. It is not near because every conceivable sector has already been exhausted. It is near because the core unification seam has ceased to be vague and has become a derived matter-sector bridge with a sharply bounded remaining strengthening programme.*

10 Derived unification theorem

We may now state the central theorem of the paper.

Theorem 10.1 (Derived DTFE bridge unification theorem)**.** *Assume:*

(i) the Darke Theory pre-geometric threshold structure, with geometry admissible only for $\Phi > \Phi_{\mathrm{crit}}$*;*

(ii) the relativistic closure–curvature bridge with post-threshold fields $(g_{\mu\nu}, \Phi, \psi)$*, positive activation, hyperbolicity, finite admissibility, and matter theorem;*

(iii) a finite-energy nontrivial DTFE branch U_B *with vacuum limit at spatial infinity;*

(iv) the standard post-threshold regularity hypotheses used in the finite-energy transfer theorem.

Then:

(a) below threshold, the theory admits a non-geometric coherence regime ordinarily described quantum mechanically;

(b) above threshold, the same theory admits an effective Lorentzian curvature regime sourced by persistent topological matter;

(c) the DTFE branch determines a canonical induced bridge matter field

$$\psi_{U_B} = v(\Phi)\, U_B \xi_0;$$

(d) this induced field has nontrivial bridge topological charge equal to the DTFE degree;

(e) this induced field has finite bridge energy;

(f) this induced field contributes a finite persistent source term to the post-threshold bridge equation;

(g) quantum gravity and general relativity therefore arise in Darke Theory as threshold-separated regimes of one and the same pre-geometric structure, connected through a derived concrete matter-sector bridge furnished by DTFE.

Proof. Item (a) is the threshold-separated consequence of the Darke Theory foundation below geometric admissibility. Item (b) follows from the relativistic closure–curvature bridge in the post-threshold regime. Item (c) is the canonical induced-field theorem. Item (d) is the exact topological transfer theorem. Item (e) is the finite-energy transfer theorem. Item (f) is the derived curvature-sourcing theorem. Thus the same pre-geometric theory supports a non-geometric near-threshold regime and an effective Lorentzian post-threshold regime linked by a derived finite-energy nontrivial topological matter sector. Once this derived bridge is in place, the unification statement follows. □

Remark 10.2. *The decisive step is the closure of the matter–geometry seam in derived form. The bridge is no longer merely structural. It now contains an explicit matter-sector realisation showing what passes from DTFE into the bridge and under which post-threshold regularity conditions that passage succeeds. For that reason the theorem is stronger than the earlier conditional transfer formulation, while remaining exact about the assumptions still in force.*

11 Conclusion

Darke Theory unifies quantum gravity and general relativity by placing both within one pre-geometric threshold structure. Below closure threshold, the theory yields the non-geometric coherence regime ordinarily described quantum mechanically. Above closure threshold, it yields an effective Lorentzian curvature regime governed by the relativistic closure–curvature bridge. The DTFE sector supplies the first concrete executable matter candidate for that bridge by providing a finite-energy nontrivial topological branch from which a canonical post-threshold matter field may be constructed.

The present paper strengthens the unification claim in one decisive way. It no longer merely isolates the matter–geometry seam. It derives it. Rather than merely asserting that DTFE is relevant to the bridge, it states the canonical induced bridge field, proves exact transfer of topological nontriviality, proves finite bridge matter energy under standard post-threshold regularity assumptions, derives the corresponding induced stress tensor, and shows that the induced DTFE sector enters the post-threshold curvature equation as a finite persistent source.

The remaining burden is therefore no longer the original vague unification seam itself, but the narrower task of strengthening, extending, or relaxing the regularity assumptions under which the present result is obtained, and of extracting further weak-field and phenomenological consequences from the sourced bridge equations.

Quantum gravity and general relativity are therefore unified in Darke Theory in the following strongest honest sense presently supported: both arise from the same underlying coherence structure, and the DTFE matter sector furnishes a derived concrete bridge by which pre-geometric closure becomes post-threshold relativistic matter and curvature source.

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